Enhanced Dysarthria Severity Detection Through Transfer Learning Using Voice Recordings

Venkata Siva Manoj A*, Velaga Lela Lakshman*, Akash Kamuju*,
P Venu*, Jyothish Lal G*
*Amrita School of Artificial Intelligence, Coimbatore, Amrita Vishwa Vidyapeetham, India
{cb.en.u4aie21074, cb.en.u4aie21073, cb.en.u4aie21021, cb.en.u4aie21044}@cb.students.amrita.edu

***Abstract*—Dysarthria, a neuromuscular communications disorder presented with impaired pronunciation, is a daunting task to identify and quantify the level of dysfunction. Through this paper, a critical study of automated dysarthria severity classification using deep learning methods will be discussed. Transfer learning is negotiated by ten different deep learning architectures on UA-Speech Corpus and TORGO datasets. The study reviews the models' performance focusing on accuracy, precision, recall and F1-scores. Experiments results prove the efficiency of the developed method for correct classifying dysarthria severity levels, and it puts forward the opportunity of the use of automated diagnostic units in the neurology clinical practice. Besides, the study presents the selection of efficient deep learning network models for the intended task and sets the baseline for the leading work in the area of dysarthria classification.**

***Index Terms*—Transfer Learning, Speech Processing, Dysarthria Severity Detection, Deep Learning**

I. Introduction

Dysarthria, a motor speech disorder in which there is difficulty in articulation as a result of muscle weakness, paralysis, or inability to coordinate the muscles involved in speech production. It happens due to tissue damage of the nervous system that controls the speech muscles which affects the ability to pronounce speech sounds clearly. It can express itself in different forms, such as slurred, unclear speech, voice changes, and inability to control pitch and tone of voice. A number of neurological disorders like amyotropic lateral sclerosis (ALS), Parkinsons disease (PD), Cerebral palsy, and trauma to the nervous system can all be caused by dysarthria [1]. The articulation difficulties resulting from Dysarthria cause the reduction of vocal message comprehension degree through the reduced intelligibility, audibility, naturalness, and efficiency of oral communication. Dysarthria disrupts multiple components of the speech production process, resulting in poor phonation, prosody, and articulation [2]. Dysarthria can cause challenges with society, thoughts of isolation, despair, along with various psychological issues [3]. As a result, if immediate assistance and early treatment are not provided, disease course management would be difficult, and the condition will progress further. Doctors may detect dysarthria primarily but it is often regarded as a costly, arduous, and time-consuming diagnostic [4].

The degree of dysarthria(severity level) should be evaluated as a critical diagnostic step, since this is of great importance for understanding the course of the underlying condition and guidance of clinical decisions regarding therapy or drugs when needed. Also the periodical speech therapy sessions are planned in accordance with it and then they are conducted. Speech-language pathologists are often trained to use intelligibility tests in conjunction with voice clinics to conduct assessments. Fortunately using such an approach can be expensive and variable [5]. The implementation of a standardised objective severity evaluation is useful in a variety of ways, including cost-effectiveness and consistency.

The use of speech analysis as a diagnostic tool bears great promises in neurology and oftentimes is linked with diagnosing complex neurological disorders like dysarthria. These disorders usually generate their own distinct speech patterns, which can be utilised as key diagnostic markers. Speech analysis is one of the most important simple and cost-effective tools for early detection and treatment planning of various neuro-disorders [6]. Medical professionals may be able to aid with early diagnosis and treatment planning by carefully examining such speech traits, which may result in an increase in patient quality of life and the ability to provide timely treatments.

To the best of the authors' knowledge, this is the first study which utilizes a transfer learning approach to categorise dysarthria severity into very low, low, medium and high levels, depending on the voice-based features and deep learning. This study presents 10 distinct deep learning architectures for classifying dysarthria severity using speech indicators. Then, a detailed comparison study is carried out to determine the most finely tuned deep learning architecture that handles the multi-class classification challenge. The key contributions of our study can be stated as follows:

- We have used a Transfer Learning method to evaluate the severity of dysarthria, achieving impressive accuracy using only voice-based features.
- Using an open-source dataset, our study provides a significant benchmark for future research in this area of study.

The subsequent sections of this paper are structured as follows: Section II contains a brief description of the literature review, Section III describes about the data used and provides a thorough overview of the methodology, Section IV shows the results, and Section V presents the conclusion and future scope.

II. Literature Survey

Many automated dysarthria severity level classification methods rely on acoustic data to identify the key aspects of dysarthric speech production [7], [8]. The inclusion of "proper choice of a deep learning architecture can ensure better performance than the conventionally used SVM classifier" in [9] is to highlight the highest classification accuracy of 96.18% and 93.24% when DNN, CNN, and LSTM were employed for analyzing UA-Speech corpus and TORGO database using MFCC coefficients and its derivatives as features, compared to SVM classifier. Additionally, particular features of speech-disorder identified from prosody, articulation, phonation, and glottal functioning were evaluated on DNN models in [10]. When investigating UA-Speech database using DNN classifier with MFCC based i-vectors excelled with 93.97% accuracy under speaker-dependent scenario by examining the usefulness of low-dimensional feature representation using subspace to yield i-vectors. In [11], by delving into integrity and validity of ten speech-language pathologist's, the speech severity of 52 amyotrophic lateral sclerosis and 20 healthy speakers was rated. Thresholds for dysarthria severity were determined by Receiver operating characteristic (ROC) curves. Difficulties arose when opting for a stratification indicator. To improve the unbiased estimation of intelligibility in speech, Mahalanobics distance based discriminant analysis classifier was developed to classify into two levels which was trained on 34 acoustic features, achieving 95% accuracy in [12]. Integrating the mappings formulated by selecting nine prominent features, results in enhancement up to 13% in accuracy. [1] uses ANN with Multitapered spectral estimation based features and audio descriptors as attributes for classifying speech severity.Examination of UA-Speech corpus and TORGO database brings about 96.44% and 98.7% accuracy.

Further in [13], utterances of 13 healthy speakers and 19 dysarthrics detected speakers from Universal Access Speech database were employed as mel spectrograms and given to a residual CNN. To recognize the salient periods in the speech, multi-head attention mechanism was incorporated. Uplift of 11.51% in accuracy was observed compared to the initial system. A cross-database approach is implemented in [14], where models were trained on UA-Speech and tested on Atlanta Motor Speech Disorders Corpus (AMSDC) divisions. To record distinctive features of dysarthric speech Teager Energy Operator (TEO) and glottal waveform were embedded, which resulted in 90% accuracy. Digital signal processing algorithms are employed for evaluating severity of features in [15]. Linear discriminant classifier and non-linear classifier are utilized for classification. Non-linear classifier surpasses linear classifier which additionally delivers bi-dimensional representation of the data. [16] merges Linear Discriminant Analysis (LDA) with Gaussian Mixture Model (GMM) and Support Vector Machine (SVM). Founded on distinguishing capacity prosodic features were involved in front end processing. Analysis on Nemours database of American dysarthric speakers yields in classification rate of 93%. In [17] discerning dysarthria and evaluating its severity was by inspecting the rhythm metrics. Accuracies of rhythm-based features by employing Random Forest, Support Vector Machine and feed-forward neural network were compared with accuracies achieved with prosodic features. Progress of 7.5% and 15% in detection and severity analysis was found for QoLT Korean dataset. Vocalized segments from two controlled speakers and two dysarthric speakers with low and intermediate range each originating from UA database were examined in [18], using breathiness features for reviewing the intelligibility of dysarthria and discriminating the type. In parallel with other studies, pronunciation is taken into account as an additional factor for extracting features in [19], thus characteristics of dysarthric speech are accurately depicted. Percentage of correct phonemes and degree of vowel distortion assessed as pronunciation features. In [20] Hierarchical cluster analysis is implemented on twenty-five-year-old children with dysarthria secondary to CP and twenty normally progressing children as a control group to predict the organic subgroup of children derived from six acoustic and perceptual speech measures to measure segmental and suprasegmental speech characteristics. [21] tried to elevate the efficacy and applicability of speech-powered tools by proposing a Residual Network-based strategy on standard Universal Access corpus, considering short duration speech segments as input features and comparison was made against CNN, GMM and Light CNN's. Dysarthria arisen from Amyotrophic Lateral Sclerosis was dealt in [22]. Data from 119 ALS and healthy speakers was employed to conduct automatic classification of vowels and fricatives at diverse degrees of severity of dysarthria and appraise the findings with manual classification, where boost in accuracy was found.

The review presents the state of the art in fully automated dysarthria severity analysis, showing the different approaches and techniques used in the researches journals. Studies have come to use the acoustic attributes, deep learning architectures, and speech assessment techniques to evaluate dysarthria severity levels correctly. Nevertheless, the ranking achieved is relatively low in terms of accuracy and scalability owing to multi-class classification and database heterogeneity Based on the foundation started by using transfer learning with deep-neural networks we intend to improve the severity detection of dysarthria. Our novel idea is based on the application of ten deep learning architectures based on transfer learning, giving us an opportunity to make an in-depth comparison of models to choose the best one for this task. In addition, our investigation is a milestone for upcoming research in dysarthria categorization which would rely on open-source datasets and standard evaluation indicators.

III. Materials and Methodology

In this section, we provide the materials and approach utilized in our work for diagnosising the severity of dysarthria using transfer learning and deep neural networks. The datasets used for this study are including the UA-Speech Corpus and the TORGO dataset, along with an overview of the methodology adopted for model training and evaluation. Furthermore,

the deep learning architectures selected for transfer learning are outlined, along with the simulation configuration and evaluation metrics used to assess model performance.

A. Data

The proposed work was validated with the Universal Access Dysarthric Speech Corpus (UA-Speech), a standard American English dysarthric database [7] and TORGO dataset [23].

1) UA Dysarthric Speech Dataset: The UA-Speech dataset contains the recordings from 13 healthy speakers and 19 dysarthria patients, but only the data from 15 patients were available. Each participant provided 155 common words, repeated three times, standing for numbers, computer orders, international radio alphabets and 100 common words from the Brown corpus, resulting in 465 common words per speaker. In addition, the corpus includes 300 different rare words per speaker, gathered through a greedy algorithm from children's novels digitized by Project Gutenberg and according to a sequence pooling criterion. Data from the sixth microphone channel, sampled at a frequency of 16 kHz, were utilized due to its superior signal-to-noise ratio. Severity levels were assigned very low, low, medium, and high based on the intelligibility ratings provided by five naive listeners for each speaker in UA-Speech.

2) Torgo: The TORGO dataset contains healthy person's 3D articulator markers and speech records acquired simultaneously, along with articulatory and acoustic data from those persons suffering from cerebral palsy or amyotrophic lateral sclerosis(ALS). It entails auditory stimuli of nonwords, short words, and phrases of known and unknown interchange, acquired using two types of microphones, an array one, and a wearable one. The data sets includes the twenty-six English digits, international radio alphabets, some common English words, and phonetically contrasting pairs to provide a wide range of speaking stimuli. The Perceptual dimensions of speech were assayed using the Frenchay Dysarthria Assessment (FDA) [24] that is a standardized assessment meant to assess various motor aspects of speech. With 16 kHz head-mounted microphone recordings, the computer-generated TORGO dataset has been collections for speech dysarthric speech studies towards understanding both acoustic and articulatory elements across different speech tasks and participant groups [23].

The severity classification for both the UA Speech and TORGO databases is outlined in table I. In the UA Speech database, severity classification is determined based on intelligibility, while in the TORGO database, it relies on the overall FDA score for dysarthric speakers as referenced in [25]. Each voice recording is then transformed to a mel-spectrogram, which is resized to 224x224 and input into Deep Learning models as shown in the figure 1.

B. Methodology

We investigated the efficiency of transfer learning with this problem. In this regard, we used 10 models as described in the following subsection, adding a classification head to each. The classification head consists of 6 linear layers with Rectified Linear Unit activations. The fine-tuning of the whole model, including the classification head, is done using the Adam optimizer with a learning rate of 0.01. We continued training all models until they reached convergence and started displaying signs of overfitting after over 150 epochs. Then, we selected the model that showed the highest validation accuracy retrospectively. All the test metrics in the study are extracted from the model whose validation accuracy is the highest. The figure 2 below represents the proposed methodology for the classification task.

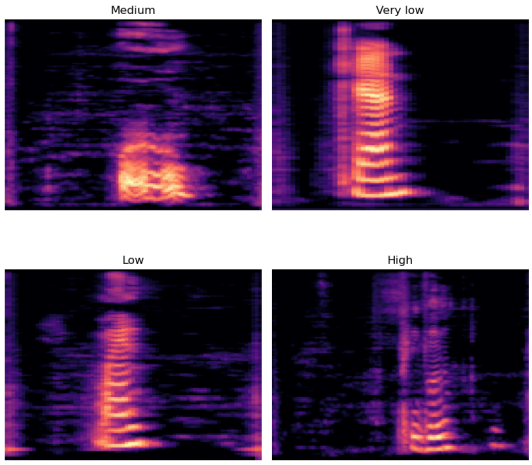


Fig. 1. Spectrograms Representing Different Dysarthria Severity Classes

TABLE I
SEVERITY CLASSIFICATION FOR UA SPEECH AND TORGO DATABASES

Severity	UA Speech	TORGO
Very Low	F05, M08, M09, M10, M14	F03, F04, M03
Low	F04, M05, M11	F01, M05
Medium	F02, M07, M16	M01, M02, M04
High	F03, M04, M12, M01	-

1) Simulation Configuration: The training process utilizes the PyTorch library, with experiments conducted on the Kaggle platform, leveraging a T4 GPU for computational assistance. Detailed descriptions of the hyperparameters utilized during training are provided in the table III-C.

C. Models

1) Vision Transformers
2) ResNet50
3) ResNet152
4) GoogleNet
5) VGG16
6) VGG19
7) DenseNet121
8) DenseNet169
9) DenseNet201
10) MobileNetV2

Vision Transformers [26], [27], introduce a new concept of image classification by using self-attention scheme instead

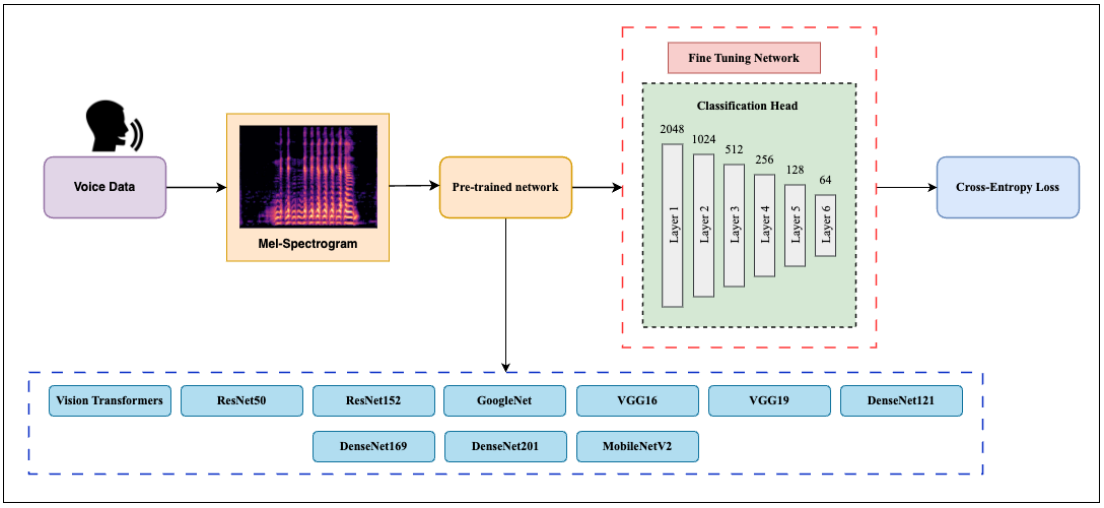


Fig. 2. Flow Diagram of the Proposed Methodology

TABLE II
TRAINING PARAMETERS

Parameter	Value
Optimizer	ADAM
Learning rate	10^{-4}
Num Epochs	150
Batch size	32

of the widely used convolutional layers. Through treating images as sequences of tokens ViTs manage to capture the global context and dependencies by doing that they literally amplify the feature extraction process. Their scalability lets adapting these methods to different resolutions in which input images are split into patches and linear embedding. This novel architecture design provides a breakthrough by presenting new computer vision, with enhanced flexibility and performances in image analysis tasks.

GoogleNet [28], is an advanced deep convolutional neural network. It is remarkable for its efficiency and effectiveness with 22 layers being used for computational efficiency and accuracy simultaneously, using inception modules with multiple kernel sizes. This new approach makes it possible for GoogleNet to capture features of different scales and scales flexibly, leading to high accuracy while reducing computation complexity. On the other hand, GoogleNet operated also, global average pooling to reduce weights and overfitting issues, thereby becoming one of the most significant architectures of this area.

ResNet50 and ResNet152 [29] are deep convolutional neural networks, frequently employed in image classification tasks. ResNet50 is a 50 layered neural network, while ResNet152 has 152 layers. Both architectures employ skip connections in order to solve the vanishing gradient problem, thereby making training of deeper networks more efficient. Such innovation leads the networks to detect hidden patterns in the images with more precision, and hence better classification performance.

VGG16 and VGG19 [30] are two convolutional neural networks, features a series of convolutional layers with 3x3 filters followed by max-pooling layers for spatial down-sampling. Although they do well in asking hierarchical features from images, the deep structures that lie behind them cause high computational costs and resource requirements for training and inference.

DenseNet121, DenseNet169, and DenseNet201 [31] architectures in convolutional neural networks are employed for classifying images. These interconnected or dense blocks enhance the model performance since each block receives input from all previous layers. These models are also differ in terms of depth, with the counting for DenseNet121 being 121 layers, DenseNet169 being 169 layers, and also DenseNet201 being 201 layers. Paradoxically, despite their rising complexity, DenseNets architectures hold a reputation for its parameter effectiveness, allowing them to attain state-of-the-art accuracy with fewer parameters when compared to the traditional architectures. By virtue of trait of using dense connections between layers, DenseNet models enable feature reuse and parameter efficient utilization, which in turn produce top performance in variety of computer vision tasks, notably image classification and object detection.

MobileNetV2 [32] is an optimised and fast convolutional neural networks with focus on the domain of image classification on the mobile devices. That reduces the size of the model and increases the model's computational efficiency with the inverted residual blocks and linear bottlenecks played its role, so it leads to more faster and more accurate classification for real-time tasks. This kind of architecture represents the next stage of development of the initial MobileNet architecture and includes improvements in the accuracy while maintaining the low computational costs and attribute of the architecture it makes it suitable for vision tasks on devices that are resource-limited, such as image classification and object detection.3 depicts the split of the dataset into training, testing, and

validation.

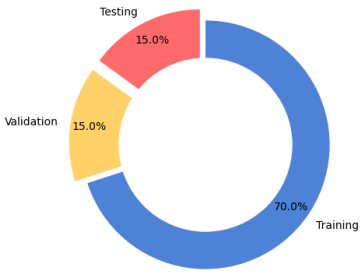


Fig. 3. Visualization of data split for training, testing and validation.

D. Evaluation Metrics

The evaluation metrics that have been used in this study include accuracy, F1-measure, precision and recall. Accuracy gives an overall idea on the correctness of predictions made by a model while F1-score provides the balance between the precision and recall, which is very significant when facing with imbalanced datasets. Precision is a measure of how good are positive predictions correctly determining true positives among all positive predictions. In contrast, recall measures the ratio of all positive instances correctly identified by a model compared to relevant instances that the model missed. Table III-D illustrates the evaluation metrics used in this work.

TABLE III
EVALUATION METRICS USED IN THIS STUDY FOR ASSESSING THE EFFECTIVENESS OF THE CONSIDERED DL MODELS.

S.No	Error metric	Formula
1	Accuracy	$\frac{TP+TN}{TN+FP+TP+FN}$
2	F1 score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
3	Precision	$\frac{TP}{TP+FP}$
4	Recall	$\frac{TP}{TP+FN}$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

IV. Experimental Results

The results for some models have been documented thus far, with additional models yet to undergo analysis.

TABLE IV
PERFORMANCE METRICS OF VARIOUS MODELS (MEAN ± STANDARD DEVIATION OF 4 RUNS) FOR UA DATASET

Model	Accuracy	F1	Precision	Recall
ViT-L-32				
ViT-B-16				
MobileNetV2				
DenseNet201				
DenseNet169				
DenseNet121				
ResNet152				
ResNet50				
GoogLeNet	0.9933	0.9933	0.9931	0.9935
VGG19	0.9913	0.9905	0.9911	0.9900
VGG16	0.9895	0.9883	0.9883	0.9882

TABLE V
PERFORMANCE METRICS OF VARIOUS MODELS (MEAN ± STANDARD DEVIATION OF 4 RUNS) FOR TORGO DATASET

Model	Accuracy	F1	Precision	Recall
ViT-L-32				
ViT-B-16				
MobileNetV2				
DenseNet201	0.9939	0.9934	0.9929	0.9939
DenseNet169	0.9970	0.9970	0.9963	0.9977
DenseNet121	0.9970	0.9970	0.9963	0.9977
ResNet152	0.9848	0.9846	0.9845	0.9847
ResNet50				
GoogLeNet	0.9848	0.9835	0.9817	0.9855
VGG19	0.9939	0.9935	0.9929	0.9942
VGG16	0.9939	0.9934	0.9929	0.9939

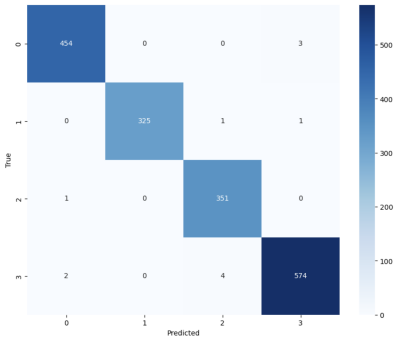


Fig. 4. Illustration of the confusion matrix of GoogleNet

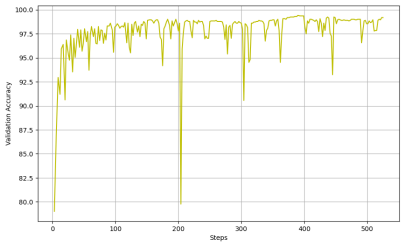


Fig. 5. Illustration of the validation accuracy plot of GoogleNet model.

V. Conclusion

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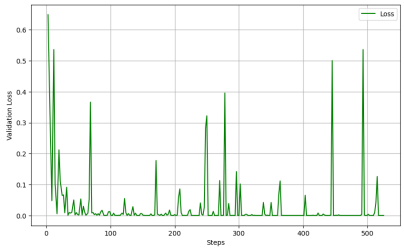


Fig. 6. Illustration of the validation loss plot of GoogleNet model.

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